Adjoint-Based Gradients for Gradient-Enhanced Gaussian Process Regression in Non-Intrusive POD Reduced-Order Models

Andrew N. Jeffery [1,*], Jean C. Ragusa[1], Shawn D. Pautz[2]

[1]Department of Nuclear Engineering, Texas A&M University, College Station, TX;
[2]Sandia National Laboratory, Albuquerque, NM

[leave space for DOI, which will be inserted by ANS]

ABSTRACT

Non-intrusive reduced-order models (ROMs) built from proper orthogonal decomposition (POD) and Gaussian process regression (GPR) replace an expensive coupled-physics solve with a cheap parameter-to-quantity-of-interest (QoI) surrogate, but a value-only GP extracts a single scalar per training solve. Gradient-enhanced GPR (GE-GPR) is a well-established remedy; the contribution here is the route by which the gradients are obtained. Holding the POD basis fixed turns each retained coefficient into a scalar QoI, so one adjoint solve per mode returns its exact derivative with respect to *every* active parameter at once, reusing the factorization already paid for by the forward solve. Gradient cost therefore scales with the retained rank R, not the parameter count p, and the snapshot derivative field is never required, which makes the recipe portable to production transport codes where adjoints are routine but derivative fields are not. The construction is verified against exact adjoint gradients in a 1D one-group diffusion sandbox, on piecewise-constant material layers and on a smooth void/Teflon interface. Combining both into one 5-parameter case at a matched budget of M=80, GE-GPR cuts median field-reconstruction error by 77% (6.8×10^{-4} vs. 3.0×10^{-3}) and lands $\sim$ 3.5× above the POD projection floor against $\sim$ 15× for standard GPR, a separation confirmed over five independent Latin hypercube designs; on the hardest single parameter it reaches standard GPR's converged accuracy with roughly an order of magnitude fewer solves. A SCEPTRE SGEMP charge-deposition problem gives a converged standard-GPR baseline ($\varepsilon_q \approx 1.15\times10^{-3}$) as the first step toward a production multi-dimensional transport solve.

Keywords: reduced-order modeling, proper orthogonal decomposition, Gaussian process regression, adjoint sensitivity, gradient-enhanced surrogate

1. INTRODUCTION

Coupled-physics solves are often too expensive to call repeatedly inside a design loop, an uncertainty-quantification study, or an optimization routine. Non-intrusive reduced-order models (ROMs) replace the expensive solver with a cheap surrogate map from parameters μ to a quantity of interest (QoI). In the standard proper orthogonal decomposition (POD) recipe, a snapshot matrix of the QoI is factored by a thin singular value decomposition (SVD) to give a low-rank basis Ψ, and a Gaussian process (GP) learns the map $\mu \mapsto \alpha(\mu)$ from parameters to the reduced coefficients of $y(\mu) \approx \Psi\alpha(\mu)$.

The limitation is that a value-only GP sees function values alone, so each expensive solve contributes a single scalar per POD mode. Supplying derivative observations is a well-established remedy: GP regression

*andrewnj03@tamu.edu

on derivative data dates to Morris, Mitchell and Ylvisaker [1], and gradient-enhanced kriging (GEK) or GPR (GE-GPR) has since been developed extensively [2, 3, 4, 5, 6, 7], with Zimmermann [8] combining it with POD. The GE-GPR estimator is therefore not claimed as new here.

The estimator is not the contribution; the route to the gradients it is fed, and what that route costs, is. The POD basis Ψ is computed once and held fixed, which turns each retained coefficient $\alpha_k(\mu) = \psi_k^\top H\phi(\mu)$ into an ordinary scalar QoI, linear in the state. Classical adjoint sensitivity analysis [9] then applies directly: one adjoint solve per retained mode returns $\partial\alpha_k/\partial\mu_j$ for *every* active parameter μ_j simultaneously, reusing the factorization of the forward system already computed for $\phi(\mu)$.

Three consequences distinguish this from the GEK literature above.

1. *Gradient cost scales with rank, not dimension.* A training sample costs R adjoint solves (R the retained rank, here 1 and 2) and yields $R \times p$ derivative observations, so adding a parameter adds no solves. Finite-difference or tangent-mode enhancement instead costs $O(p)$ extra solves per sample, which is what limits GEK in high dimension.
2. *No snapshot derivative fields are needed.* Gradient-enhanced POD surrogates in the vein of Ref. [8] require the snapshot derivative $\partial y/\partial\mu \in \mathbb{R}^{N_y}$, which a production transport code generally does not expose; the adjoint delivers the R coefficient derivatives directly and never forms it. This matters where adjoint solves are routine machinery (shielding, detector response, perturbation theory) but derivative fields are not.
3. *The gradients are exact.* Because α_k is linear in ϕ, the sensitivity is algebraically exact to solver tolerance, so any sample-efficiency gain reflects the information content of the gradients, not finite-difference step-size tuning.

A one-group, 1D diffusion solver serves as a sandbox for scoping the recipe rather than as the paper's object of interest: its adjoint sensitivities can be verified in closed form and by finite differences, so every sample-efficiency claim is checked exactly before the recipe reaches a code where it cannot be. The pipeline (forward solve → POD → adjoint → GE-GPR) is specific neither to the diffusion solver nor to one spatial dimension.

2. PROBLEM SETTING AND BASELINE ROM

The construction asks little of the high-fidelity solver: only that a solve produce a discrete state from a parametrized linear (or linearized) system, and that the QoI be a linear functional of that state. *Nothing* in Secs. 3 and 4 depends on spatial dimension, physics package, or discretization. With $\mu \in \mathbb{R}^p$ the active parameters, a solve returns the state $\phi(\mu) \in \mathbb{R}^N$ from

$$A(\mu)\,\phi(\mu) = s(\mu), \tag{1}$$

with $A(\mu)$ the discrete operator and $s(\mu)$ the assembled right-hand side. The QoI is a field-valued linear observation of that state,

$$y(\mu) = H\,\phi(\mu) \;\in\; \mathbb{R}^{N_y}, \tag{2}$$

where H is any fixed linear operator: a spatial mask, a detector response, a surface tally. Snapshots at M_{POD} training parameters form $Y \in \mathbb{R}^{N_y \times M_{\mathrm{POD}}}$, factored by a thin SVD. Retaining R left singular vectors at an energy cutoff η gives the POD basis $\Psi \in \mathbb{R}^{N_y \times R}$, whose orthonormal columns yield the reduced coefficients by projection:

$$y(\mu) \;\approx\; \Psi\,\alpha(\mu), \qquad \alpha(\mu) = \Psi^\top y(\mu) \;\in\; \mathbb{R}^R. \tag{3}$$

The baseline surrogate is one independent ARD-RBF-kernel GP per retained mode (scikit-learn's `GaussianProcessRegressor`), evaluated as $\hat{y}(\mu) = \Psi\hat{\alpha}(\mu)$. Results are the relative field-reconstruction error on a

held-out test set, read against the POD projection floor:

$$\varepsilon_y(\mu) = \frac{\|\hat{y}(\mu) - y(\mu)\|_2}{\|y(\mu)\|_2}, \tag{4}$$

$$\varepsilon_{\text{proj}}(\mu) = \frac{\|\Psi\Psi^\top y(\mu) - y(\mu)\|_2}{\|y(\mu)\|_2}. \tag{5}$$

The floor $\varepsilon_{\text{proj}}$ is the error of rank-R truncation alone, so it is the best any surrogate on this basis could do, and every regression gain below is read against it.

2.1. Demonstration Platforms

The method is demonstrated on two solvers at opposite ends of the cost and verifiability scale:

1. a 1D, one-group neutron diffusion *sandbox*: cheap enough to sweep exhaustively and simple enough that every adjoint sensitivity can be checked in closed form and by finite differences (Secs. 2.2, 5.1-5.4);
2. SCEPTRE, Sandia's export-controlled multi-dimensional coupled electron-photon transport code, on a full multi-D SGEMP charge-deposition solve (Sec. 5.5): a production code whose gradients cannot be verified by hand.

The sandbox carries the GE-GPR comparison; SCEPTRE currently establishes the value-only baseline, with its adjoint output targeted for the December review. Because Eqs. (1)-(3) are all the adjoint construction of Sec. 3 uses, moving between the two changes nothing in the method, only the routines supplying $\partial A/\partial \mu_j$ and $\partial s/\partial \mu_j$.

2.2. The 1D Diffusion Sandbox

The sandbox solves a 1D, one-group neutron diffusion problem,

$$-\frac{d}{dx}\left[D(x)\frac{d\phi}{dx}\right] + \Sigma_a(x)\,\phi(x) = q(x), \qquad x \in (0, L), \tag{6}$$

with Dirichlet boundary conditions, discretized by a cell-centered, second-order finite-volume scheme on a uniform N-cell mesh into the sparse system of Eq. (1). The operator H of Eq. (2) masks a fixed window of cells, so $y(\mu)$ is the flux restricted to that window. Two parameterizations of μ give the two test cases carried through the paper:

- **Material layers:** μ is the per-layer diffusion coefficient, absorption cross section, and source, $(D_m, \Sigma_{a,m}, q_m)$, piecewise constant over N_{mat} layers.
- **Void/Teflon geometry:** a symmetric slab (Shield | Teflon | Void | Core | Core | Void | Teflon | Shield) in which only the Void/Teflon interface may move. A hard boundary is not differentiable in its own location, since moving it changes which cell it falls in, so at that interface the step becomes a logistic switch

  $$\sigma(x; a, b) = \left[1 + e^{-a(x-b)}\right]^{-1}, \tag{7}$$

  with b the interface location ($\sigma(b) = 0.5$ for any a) and a the steepness. The Core/Void and Teflon/Shield boundaries stay hard steps, since they do not move here.

All three coefficients are cell-wise arrays at cell centers x_i, with no sub-cell averaging: piecewise constant in the material-layer case, and in the geometry case blended across the moving interface by the same switch,

$$v(x_i) = v_{\text{void}} + (v_{\text{Teflon}} - v_{\text{void}})\,\sigma(r_i; a, b)\,, \qquad r_i = |x_i - L/2|, \tag{8}$$

for $v \in \{D, \Sigma_a, q\}$ within the shell $r_i \in [r_{\text{core}}, r_{\text{core}} + w_{\text{shell}}]$, and fixed core and shield values outside it. One switch for all three keeps a single (a, b) pair, so the same two derivatives $\partial\sigma/\partial a$ and $\partial\sigma/\partial b$ propagate to every coefficient field.

In the assembled system Σ_a and q enter cell-locally, while D enters only through the interior-face conductances

$$\beta_{i+1/2} = \frac{2\, d_i d_{i+1}}{d_i + d_{i+1}}, \qquad d_i = \frac{D_i}{\Delta x_i}, \tag{9}$$

a harmonic mean of half-cell conductances. That distinction is what makes the D sensitivity of Sec. 3 more involved than the other two.

3. ADJOINT SENSITIVITIES OF THE POD COEFFICIENTS

Each POD coefficient is a scalar QoI that is linear in the state,

$$\alpha_k(\mu) = \psi_k^\top y(\mu) = \psi_k^\top H \phi(\mu), \tag{10}$$

so the classical adjoint construction [9] applies without modification. Differentiating the residual $\mathcal{R}(\phi, \mu) = A(\mu)\phi - s(\mu) = 0$ with respect to μ_j and eliminating $\partial\phi/\partial\mu_j$ with an adjoint variable gives the central relation of this paper,

$$\boxed{\frac{\partial \alpha_k}{\partial \mu_j} = \lambda_k^\top \left(\frac{\partial s}{\partial \mu_j} - \frac{\partial A}{\partial \mu_j}\, \phi \right), \qquad A^\top \lambda_k = H^\top \psi_k.} \tag{11}$$

The adjoint system is solved once per retained mode k, reusing the factorization of A (transposed) already computed for the forward solve. The right-hand contraction is then one matrix-vector product per parameter, so all p partials of mode k follow from that single solve: a training sample costs R adjoint solves and yields $R \times p$ derivative observations, independent of p.

The same λ_k also reproduces the coefficient itself,

$$\alpha_k = \left(A^\top \lambda_k\right)^\top \phi = \lambda_k^\top s, \tag{12}$$

which is used below as a verification identity. Everything case-specific then reduces to supplying $\partial A/\partial\mu_j$ and $\partial s/\partial\mu_j$.

Material-layer case. For the per-layer source and absorption, Eq. (11) collapses to closed forms, since $\partial s/\partial q_m$ and $\partial A/\partial \Sigma_{a,m}$ are cell-local and supported only on layer m. With Δx_i the cell width, layer(i) the layer index of cell i, and the sums restricted to interior cells, where no Dirichlet row overwrites the residual:

$$\frac{\partial \alpha_k}{\partial q_m} = \sum_{i \,\in\, \text{layer}\, m,\ \text{interior}} (\lambda_k)_i \, \Delta x_i, \tag{13}$$

$$\frac{\partial \alpha_k}{\partial \Sigma_{a,m}} = -\sum_{i \,\in\, \text{layer}\, m,\ \text{interior}} (\lambda_k)_i \, \phi_i \, \Delta x_i. \tag{14}$$

The diffusion-coefficient sensitivity takes more work, since D enters only through the face conductances of Eq. (9). Differentiating that harmonic mean gives

$$\frac{\partial \beta_{i+1/2}}{\partial D_i} = \frac{2\, d_{i+1}^2}{(d_i + d_{i+1})^2\, \Delta x_i}, \tag{15}$$

and symmetrically for D_{i+1}. Assembling these face terms into $\partial A/\partial D_m$ over the faces bordering layer m and contracting as in Eq. (11) gives

$$\frac{\partial \alpha_k}{\partial D_m} = -\lambda_k^\top \frac{\partial A}{\partial D_m}\,\phi. \tag{16}$$

Geometry case. Differentiating the blend of Eq. (8) through the logistic switch gives smooth per-cell "direction vectors" $\partial(D, \Sigma_a, q)/\partial a$ and $\partial(D, \Sigma_a, q)/\partial b$ in place of the layer indicator above; assembling $\partial A/\partial \mu_j$ and $\partial s/\partial \mu_j$ from them, Eq. (11) applies unchanged. Because that construction is generic to any per-cell derivative field, the same code extends without modification to material parameters promoted from fixed constants to active dimensions (Sec. 5.3).

Verification. Every sensitivity above is checked three ways, identically in both test cases:

1. an *identity check*, comparing $\alpha_k = \lambda_k^\top s$ from Eq. (12) against the direct projection $\alpha_k = \psi_k^\top y$, which must agree to solver tolerance;
2. a *finite-difference check* against a centered difference over a small parameter perturbation;
3. a *1D sweep* of $\alpha_k(\mu)$ with the adjoint-predicted tangent line overlaid at several expansion points.

The first two run on every case below (identity error $\sim 10^{-11}$, finite-difference agreement $\sim 10^{-5}$ relative); the sweep is a per-parameter diagnostic.

4. GRADIENT-ENHANCED GAUSSIAN PROCESS REGRESSION

The estimator is standard [1, 3]; this section fixes notation and the implementation choices the results depend on. A GP is closed under differentiation, so if $\alpha_k(\mu) \sim \mathcal{GP}(0, k(\mu, \mu'))$ the joint distribution of α_k and $\partial\alpha_k/\partial\mu$ is itself Gaussian, with covariance blocks obtained by differentiating the kernel. Training on values and gradients together is therefore not an approximation bolted onto a value-only GP; it is the same model, conditioned on more of what it already implies. Feeding gradients in as extra features would discard the cross-covariance $\mathrm{Cov}(\alpha_k, \partial\alpha_k/\partial\mu_m)$ that makes a gradient informative about *nearby* function values. Stacking values and gradients into one observation vector,

$$y_{\mathrm{aug}} = \begin{bmatrix} \alpha_k \\ \partial\alpha_k/\partial\mu \end{bmatrix}, \tag{17}$$

under one augmented kernel is what preserves it.

For an ARD-RBF kernel $k(x, x') = \sigma_f^2 \exp\!\left(-\tfrac{1}{2}\sum_j r_j^2/\ell_j^2\right)$ with $r_j = x_j - x'_j$, the value-value block is k itself and the other two are

$$\mathrm{Cov}\left(f(x), \frac{\partial f}{\partial x'_m}(x')\right) = k(x, x') \cdot \frac{r_m}{\ell_m^2}, \tag{18}$$

$$\mathrm{Cov}\left(\frac{\partial f}{\partial x_m}(x), \frac{\partial f}{\partial x'_n}(x')\right) = k(x, x') \cdot \left[\frac{\delta_{mn}}{\ell_m^2} - \frac{r_m r_n}{\ell_m^2 \ell_n^2}\right], \tag{19}$$

assembled into the block Gram matrix

$$K_{\mathrm{aug}} = \begin{bmatrix} K_{ff} + \sigma_{nf}^2 I & K_{fd} \\ K_{fd}^\top & K_{dd} + \sigma_{nd}^2 I \end{bmatrix}, \tag{20}$$

of size $M(1+p_{\mathrm{act}})$ for M training points and p_{act} active gradient dimensions, fit by maximizing the augmented marginal likelihood. The same hyperparameters must now explain both values and slopes, which is what lets

a handful of gradients pin down length scales that would otherwise need many more function evaluations. The cost is a Cholesky factorization of

$$O\Big([M\,(1+p_{\text{act}})]^3\Big), \tag{21}$$

and K_{aug} can become ill-conditioned when length scales shrink relative to the training-point spacing, the gradient-gradient block scaling as $1/\ell^2$. Both bound how far the method scales in p_{act} (Sec. 6).

Scaling, and the chain rule it forces. Inputs are min-max normalized to $[0,1]^p$ and each mode's coefficients standardized, $z_k = (\alpha_k - \bar{\alpha}_k)/\sigma_{\alpha_k}$. The adjoint gradients are computed in physical units and must be transformed consistently into that space before use,

$$\frac{\partial z_k}{\partial x_j} = \frac{\partial \alpha_k}{\partial \mu_j} \cdot \frac{\mu_j^{\max} - \mu_j^{\min}}{\sigma_{\alpha_k}}, \tag{22}$$

with $x_j = (\mu_j - \mu_j^{\min})/(\mu_j^{\max} - \mu_j^{\min})$. This is a silent failure mode worth stating: an inconsistently scaled gradient is still a valid-looking number, and a GE-GP trained on one will fit, report a finite marginal likelihood, and simply predict badly. A value-only GP has no such coupling and gives no warning.

Hyperparameters and conditioning. The terms σ_{nf} and σ_{nd} in K_{aug} are independent observation-noise standard deviations on the standardized values and on the scaled derivatives of Eq. (22); keeping them separate lets the fit express that values and slopes carry different effective noise once scaled. Both are learned alongside $(\sigma_f, \ell_1, \ldots, \ell_{p_{\text{act}}})$ by maximizing the augmented log marginal likelihood with L-BFGS-B in log space, initialized at 10^{-3} and bounded to $[e^{-10}, 1]$, with $\log \sigma_f$ and $\log \ell_j$ in $[-5, 5]$; a separate *fixed* jitter of 10^{-6} conditions the Cholesky. The value-only baseline uses the matching scikit-learn kernel (constant × ARD-RBF + white noise, variance bounded below at 10^{-6}) on the same targets, so any accuracy difference isolates the effect of adding gradients.

Since scikit-learn cannot take derivative observations, the implementation (`GEGaussianProcess`) is our own: an enabling detail rather than a contribution, reused unmodified across every case below.

5. RESULTS

Every diffusion case uses a uniform $N = 200$-cell mesh, a QoI window $x \in [3, 7]$, an energy cutoff $\eta = 99.999\%$ setting the POD rank R, and Latin hypercube (LHS) sampling of μ, so results sit on a directly comparable footing.

Two sample counts must be kept distinct. M_{POD} snapshots build the basis Ψ once, by thin SVD, and it is then frozen; M of those same snapshots (the first M of the design, so increasing M nests) fit the surrogate, with both surrogates seeing the same M points. A disjoint set of M_{test} snapshots is held out and enters neither the basis nor the fit, while the remaining $M_{\text{POD}} - M$ contribute to the basis only and are *not* a test set.

Every learning curve is therefore a *fixed-basis regression sample-efficiency study*: it measures how many solves the regression stage needs on an already-converged basis, and does not charge basis construction to either surrogate. Table I gives the per-case budgets and results against each case's floor, Eq. (5); the combined case's smaller M is a cost decision, since at $p_{\text{act}} = 5$ the augmented Gram matrix is already 6× wider at equal M.

5.1. Material-Layer Case

Each per-layer parameter ($D \in [0.5, 2.0]$, $\Sigma_a \in [0.05, 1.5]$, $q \in [0.0, 2.0]$, all others fixed) was run as an independent single-parameter study, so $\alpha_k(\mu)$ is a genuine 1D curve and the only question is whether the

adjoint gradient helps fit it. Table I collects the converged result for every diffusion case.

Table I. Sample budgets and median held-out ε_y per diffusion case, with each case's projection floor in parentheses. M is the matched regression budget (largest run for that case), R the rank retained at $\eta = 99.999\%$. Rows marked † are at their floor for both surrogates, so their improvement entry is uninformative.

Active parameter(s)		p_{act}	M_{POD}	M_{test}	M	R	Std GPR	GE-GPR	improvement
Material layers, one at a time									
D	$(\varepsilon_{\text{proj}} = 3.03 \times 10^{-5})^{\dagger}$	1	500	100	250	2	3.06×10^{-5}	3.03×10^{-5}	1.0%
Σ_a	(1.05×10^{-4})	1	500	100	250	2	1.58×10^{-4}	1.05×10^{-4}	33.6%
q	(2.72×10^{-7})	1	500	100	250	1	3.00×10^{-5}	1.38×10^{-6}	95.4%
Void/Teflon geometry									
void_thickness	$(8.84 \times 10^{-4})^{\dagger}$	1	300	60	250	2	8.84×10^{-4}	8.84×10^{-4}	0.0%
steepness	$(5.30 \times 10^{-4})^{\dagger}$	1	300	60	250	1	5.30×10^{-4}	5.30×10^{-4}	0.0%
both	$(9.32 \times 10^{-4})^{\dagger}$	2	300	60	100	2	9.33×10^{-4}	9.32×10^{-4}	0.1%
Combined									
all five	(1.96×10^{-4})	5	300	60	80	2	2.95×10^{-3}	6.83×10^{-4}	76.8%

The strongest improvement is for the source term q, a 95.4% reduction. GE-GPR reaches $\varepsilon_y \approx 4.8 \times 10^{-6}$ at M=10, already better than standard GPR's converged M=250 value of 3.0×10^{-5}, so it matches or beats standard GPR with roughly an order of magnitude fewer solves. Standard GPR stays above 3×10^{-5} across the whole budget, well short of that case's 2.7×10^{-7} floor; GE-GPR closes most of the remaining distance.

The $\varepsilon_{\text{proj}}$ column identifies the apparent $\sim 3 \times 10^{-5}$ floor in the D row as that case's rank-2 projection floor, not a property of either regressor, of the noise kernel (whose optimized level sits well below it), or of a hyperparameter bound. Both surrogates sit at it, so the 1.0% "improvement" measures standard GPR's residual gap rather than what gradients are worth; the same holds for Σ_a. The converged number thus understates the D case, where gradients help early but not asymptotically: GE-GPR is at the floor from M=10 onward, while standard GPR needs $M \approx 200$ (5.4×10^{-5} at M=10, 3.4×10^{-5} at M=50). That $\partial \alpha_k / \partial D_m$ is least informative is consistent with Sec. 3: it is the one sensitivity that is not a simple linear functional of λ_k.

5.2. Geometry Case

With only the interface location b (void_thickness $\in [0.001, 0.25]$) active and steepness fixed at a=200, mode 1 carries 99.99% of the SVD energy and GE-GPR shows essentially *no* benefit: the map is smooth and the design dense enough that a value-only GP already resolves it to the projection floor. Steepness alone gives the same null result, and with both active the surrogates reach the same median error at matched M=100. All three geometry rows sit at their floor, so the zeros are *saturation rather than failure*.

The two-parameter row nonetheless hides faster early convergence: GE-GPR settles at the floor by $M \approx 30$ to 40, while standard GPR does not close the gap until $M \approx 75$ to 100 (at M=10 the two are at 1.32 and 1.37×10^{-3}, both noisy below $M \approx 20$). Gradients here buy *faster convergence to the POD floor*, not a lower floor, a distinct mechanism from the combined case.



Figure 1. Combined 5-parameter learning curves. *Left*: one LHS design, bands the 75th-percentile error. *Right*: five independent designs, lines the median and bands the full design-to-design range.

5.3. Combined Case

All 5 parameters (void_thickness, steepness, and the Teflon shell's D, Σ_a and q) are active simultaneously in one shared forward/adjoint/GE-GPR pipeline, where one adjoint solve per mode still gives all 5 partials at once. At matched M=80, GE-GPR reaches a median $\varepsilon_y = 6.83 \times 10^{-4}$ against standard GPR's 2.95×10^{-3}, a 76.8% reduction, landing $\sim 3.5\times$ above the floor ($\varepsilon_{\text{proj}} = 1.96 \times 10^{-4}$) versus $\sim 15\times$. Unlike the one-at-a-time cases both surrogates stay well above the floor, so the comparison is resolvable at every budget and the advantage is a persistent gap rather than a head start that closes.

Figure 1 shows the mechanism: GE-GPR matches standard GPR's converged M=80 result at $M\approx30$ (1.61×10^{-3}), a factor ~ 2.7 fewer solves, and the curves separate from the smallest budget onward. At $p_{\text{act}} = 5$ and $R = 2$, two extra solves per sample, each reusing the factorization that produced $\phi(\mu)$, deliver ten derivative observations, while a value-only GP needs proportionally more samples. The price is paid at fit time, in the $M(1 + p_{\text{act}})$-sized kernel.

5.4. Repeatability Across Independent Designs

Nested prefixes of one LHS design confound the method with the design, since a favorable early prefix can advantage one surrogate. The combined case was therefore repeated over five independent designs (fresh train/test pair each, identical pipeline and budgets otherwise), Fig. 1 (right). The median improvement stays between 83% and 88% at every budget; at M=80 it is 85.1% (range 72.6-89.0%), with GE-GPR at 6.30×10^{-4} (range 4.92-6.83×10^{-4}) against standard GPR's 4.17×10^{-3} (range 2.30-5.70×10^{-3}). Standard GPR's design-to-design spread is roughly 2.5× wider, so the single-design 76.8% sits inside this range rather than being a favorable outlier.

5.5. SCEPTRE Secondary Problem

Unlike the sandbox, SCEPTRE is a real, export-controlled code solving a full multi-dimensional coupled electron-photon transport problem. Its QoI, the net SGEMP direct-drive charge $q_{\text{net}}(\mu)$, is scalar, so no POD stage is needed and the reported error is

$$\varepsilon_q = \frac{\|\hat{q}_{\text{net}} - q_{\text{net}}\|_2}{\|q_{\text{net}}\|_2}. \tag{23}$$

Its adjoint sources exist but are deliberately not yet wired in, so every result here uses standard GPR only and sets the baseline the extension will be measured against.

On a 1000-case void-thickness sweep (monoenergetic source, 80/20 split), ε_q falls from 1.65×10^{-3} at

M=10 to 1.25×10^{-3} by M=100, then flattens at $\approx 1.15 \times 10^{-3}$ from $M \approx 200$, consistent with q_{net} being almost linear in void thickness. This is the first held-out accuracy result for a SCEPTRE-based QoI in this line of work; an almost-linear 1D map is where gradients helped least above, so a large GE-GPR gain should not be assumed.

The 1000-case void/dielectric/shield grid (10^3 factorial, all three thicknesses active, 800/200 split) behaves differently: ε_q starts at 3.32×10^{-3}, roughly double, and is still declining at M=800 (1.53×10^{-3} at M=200, 1.38×10^{-3} at M=800) where the single-parameter case plateaued at $M \approx 200$. The slower convergence and higher error at every matched M mark this as a genuinely harder map, making it, not the near-linear sweep, the more promising target for the extension.

Table II. Held-out ε_q ($\times 10^{-3}$) for the two SCEPTRE SGEMP datasets, standard GPR only, at matched training-set sizes. $M_{\max}$ is the largest budget each split allows (800).

Dataset	M=10	50	100	200	400	$M_{\max}$
Void thickness only (1000 cases, p=1)	1.65	1.41	1.25	1.15	1.17	1.15
Void/dielectric/shield (1000 cases, p=3)	3.32	1.62	1.53	1.53	1.47	1.38

6. DISCUSSION

Read against the projection floor rather than against each other, the results fall into three regimes.

1. *Persistent gain.* Where the map is hard to learn from values alone and both surrogates sit well above the floor, gradients cut error substantially and the gap persists (q at 95.4%, the combined case at 76.8%).
2. *Faster convergence only.* Where the map is easy but the design is not yet dense, gradients buy speed rather than accuracy, both surrogates ending at the same floor (D, and the two-parameter geometry case).
3. *Saturated.* Where a modest design already saturates the basis, as in the single-parameter geometry rows, nothing is left to gain and the 0% entries are not evidence against the method.

This is also the principal caveat. Because three of the five one-at-a-time cases are floor-limited, "the relative benefit grows with p_{act}" is not established here: p_{act} and headroom-above-floor move together across these cases, and the present design cannot separate them. What the combined case does establish is that the adjoint route stays economical as p_{act} grows, gradient cost being tied to R rather than p. The augmented kernel's cubic factorization in $M(1 + p_{\text{act}})$, Eq. (21), is what bounds it, and is why that case runs at $M = 80$.

7. CONCLUSIONS AND FUTURE WORK

GE-GPR is established prior art. The contribution here is the adjoint route to the gradients it needs, in a form suited to nuclear-engineering solvers: freeze the POD basis, treat each retained coefficient as a scalar QoI linear in the state, and obtain its derivatives with respect to all active parameters from one adjoint solve per mode, Eq. (11). This combines two properties the surrounding GEK literature does not:

1. *Cost scales with rank, not dimension.* Gradient cost follows the retained rank R, typically one or two, rather than the parameter count p, so adding parameters adds derivative observations without adding solves.
2. *No snapshot derivative fields.* $\partial y / \partial \mu$ is never required, which makes the recipe portable to a production

transport code that exposes adjoint solves but not derivative fields.

Verified against exact adjoint gradients in the diffusion sandbox, the surrogate cut median ε_y by 76.8% on a combined 5-parameter problem at matched budget, strengthening to 85.1% (range 72.6-89.0%) across five independent LHS designs; on the source term the saving was roughly an order of magnitude in solves. Two real SCEPTRE datasets (Table II) gave this work its first held-out baselines outside the sandbox; the harder 3-parameter grid is the target for the extension.

Future work follows from the limitations above:

1. separating p_{act} from headroom-above-floor by holding the projection floor fixed while varying dimension;
2. controlling the augmented kernel's cubic cost with sparse or weighted GEK variants [4, 5];
3. a multi-output GP in place of one GP per mode;
4. charging basis and regression to one budget to measure end-to-end ROM cost;
5. for December, wiring SCEPTRE's adjoint into GE-GPR and running the full pipeline on a multi-dimensional transport solve.

ACKNOWLEDGEMENTS

This work was supported by Sandia National Laboratories under Contract Agreement 2565183.

REFERENCES

[1] M. D. Morris, T. J. Mitchell, and D. Ylvisaker. "Bayesian Design and Analysis of Computer Experiments: Use of Derivatives in Surface Prediction." *Technometrics*, **volume 35**(3), pp. 243–255 (1993).

[2] J. Backhaus, M. Aulich, C. Frey, T. Lengyel, and C. Voss. "Gradient Enhanced Surrogate Models Based on Adjoint CFD Methods for the Design of a Counter Rotating Turbofan." In *Proceedings of ASME Turbo Expo 2012* (2012).

[3] S. Ulaganathan, I. Couckuyt, T. Dhaene, J. Degroote, and E. Laermans. "Performance study of gradient-enhanced Kriging." *Engineering with Computers*, **volume 32**(1), pp. 15–34 (2016).

[4] Z.-H. Han, Y. Zhang, C.-X. Song, and K.-S. Zhang. "Weighted Gradient-Enhanced Kriging for High-Dimensional Surrogate Modeling and Design Optimization." *AIAA Journal*, **volume 55**(12), pp. 4330–4346 (2017).

[5] M. A. Bouhlel and J. R. R. A. Martins. "Gradient-enhanced kriging for high-dimensional problems." *Engineering with Computers*, **volume 35**(1), pp. 157–173 (2019).

[6] P. Semler and M. Weiser. "Adaptive Gradient Enhanced Gaussian Process Surrogates for Inverse Problems." (2024). URL https://arxiv.org/abs/2404.01864.

[7] A. S. Booth. "Gradient-enhancement and Gradient Predictions for Deep Gaussian Process Modeling of Expensive Computer Experiments." (2025). URL https://arxiv.org/abs/2512.18066.

[8] R. Zimmermann. "Gradient-enhanced surrogate modeling based on proper orthogonal decomposition." *Journal of Computational and Applied Mathematics*, **volume 237**(1), pp. 403–418 (2013).

[9] D. G. Cacuci. "Sensitivity theory for nonlinear systems. I. Nonlinear functional analysis approach." *Journal of Mathematical Physics*, **volume 22**(12), pp. 2794–2802 (1981).